Date:

Assignment No : 1

Topic : Number Sys conversion

Subject : Assembly language

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Number Conversition

Example:.. convert binary number (10111011001110)₂ into ( )₁₀?

Step : 2 Write the Place value.

Binary Digit: 1 0 1 1 1 0 1 1 0 0 1 1 1 0

Place value: 2^{13} 2^{12} 2^{11} 2^{10} 2^{9} 2^{8} 2^{7} 2^{6} 2^{5} 2^{4} 2^{3} 2^{2} 2^{1} 2^{0}

Power of 2: 2048 1024 512 256 128 64 32 16 8 4 2 1
4096

$(1 \times 4096) + (0 \times 2048) + (1 \times 1024) + (1 \times 512)$
$+ (1 \times 256) + (0 \times 128) + (1 \times 64) + (1 \times 32)$
$+ (0 \times 16) + (0 \times 8) + (1 \times 4) + (1 \times 2) +$
(0×1)

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4096 + 1024 + 512 + 256

+ 64 + 32 + ~~8~~ + 4 + 2

Ans

5990

~~Example~~ Convert $(?)_{10}$ into

$(???)_2$ $(1290)_{10}$ $(?)_2$

1290	2	645	R = 0
~~322~~ 645	2	322.5	R = 1
322	2	161	R = 0
161	2	80.5	R = 1
80	2	40	R = 0
40	2	20	R = 0
20	2	10	R = 0
10	2	5	R = 0
5	2	2.5	R = 1
2	2	1	R = 0
1	2	0.5	R = 1

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The Binary of $(1290)_{10}$ => is

$(10100001010)_2$ Ans

Hexdecimal

convert $(11001101)_2$ $(?)_{16}$

Make Pair 4 bit of

0001 0011 0101

8 4 2 1	8 4 2 1	8 4 2 1
0 0 0 1	1 0 0 1	1 1 0 1

Ans 1 9 D

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OCTAL

To ~~Binary~~ Decimal

$(645)_8 = (?)_{10}$

$6 \times 8^2 = 384 + 32 + 5$

4×8

5×8^0

$(421)_{10}$ Ans

Decimal To ~~OCTAL~~ Hexa.

$(421)_{10}$

421	16	5
26	16	10
1	16	1

$(1A5)_{16}$

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Hexadecimal

Hexa $(2A7)_{16} \Rightarrow (?)_2$

2		0 0 1 0
A	10	1 0 1 0
7		0 1 1 1

$(0\ 0\ 1\ 0\ 1\ 0\ 1\ 0\ 0\ 1\ 1\ 1)_2$

OCTAL

OCTAL to Hexa.

		4 2 1	
5	=	1 0 1	
7	=	1 1 1	$(1\ 0\ 1\ 1\ 1\ 1\ 0\ 1\ 0)_2$
2	=	0 1 0	

0001 0111 1010

1 7 A Ans